MATH 423: Lecture 4

Portfolio Management

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Portfolio Management: Case of two securities

Simple case: two stocks and two time spots.

	Stock # 1		Stock # 2	
	Position	Price	Position	Price
$t = 0$		$S_1(0)$		$S_2(0)$
	x_1		x_2	
$t = 1$		$S_1(1)$		$S_2(1)$

Def 4.1

We define the weights of two stocks (in the portfolio) by

$$w_1 = \frac{x_1 S_1(0)}{V(0)}, \qquad w_2 = \frac{x_2 S_2(0)}{V(0)}.$$

I.e., w_i denotes the portion of our initial wealth invested in Stock i.

Rmk 4.1

- $V(0) = x_1 S_1(0) + x_2 S_2(0) =$ initial wealth of the portfolio.
- $x_i S_i(0) = w_i V(0)$ is the amount invested in stock i today.
- $w_1 + w_2 = 1$.
- $w_1, w_2 \in \mathbb{R}$ could be zero or negative.

Eg 4.1

Suppose $V(0) = 100$, $w_1 = 30\%$, $S_1(0) = 55$, $S_2(0) = 42$. If the price of stock 1 drops to $S_1(1) = 45$, then what is the minimal value for $S_2(1)$ to reach to guarantee a return of at least 10% on the portfolio.

Let us initially compute our positions on each stock:

$$w_i = \frac{x_i S_i(0)}{V(0)} \Rightarrow x_i = \frac{w_i V(0)}{S_i(0)} \Rightarrow x_1 = \frac{30}{55} \;\&\; x_2 = \frac{70}{42}.$$

Now we want

$$\begin{aligned} K_V \geq 10\% &\Leftrightarrow \frac{V(1) - V(0)}{V(0)} \geq 10\% \\ &\Leftrightarrow \frac{x_1 S_1(1) + x_2 S_2(1) - V(0)}{V(0)} \geq 10\% \\ &\Leftrightarrow S_2(1) \geq \frac{1.1 \cdot V(0) - x_1 S_1(1)}{x_2} = 51.\overline{27}. \end{aligned}$$

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Prop 4.1

For a portfolio consisting of two securities,

$$K_V = w_1 K_1 + w_2 K_2,$$

where $K_V := \frac{V(1)-V(0)}{V(0)}$, $K_i := \frac{S_i(1)-S_i(0)}{S_i(0)}$, $i = 1, 2$ are the returns of the portfolio and of stock i over time period $[0, 1]$.

Proof: Let us initially compute $V(1)$ in terms of the returns K_i:

$$V(1) = x_1 S_1(1) + x_2 S_2(1) = x_1 S_1(0)(1 + K_1(1)) + x_2 S_2(0)(1 + K_2(1)),$$
$$V(0) = x_1 S_1(0) + x_2 S_2(0).$$

Therefore,

$$\begin{aligned} K_V &= \frac{V(1) - V(0)}{V(0)} = \frac{x_1 S_1(0) K_1(1) + x_2 S_2(0) K_2(1)}{V(0)} \\ &= \frac{w_1 V(0) K_1(1) + w_2 V(0) K_2(1)}{V(0)} = w_1 K_1(1) + w_2 K_2(1). \end{aligned}$$

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Prop 4.2

For a portfolio consisting of two securities,

$$e^{k_V} = w_1 e^{k_1} + w_2 e^{k_2},$$

where $k_V := \ln \frac{V(1)}{V(0)}$, $k_i := \ln \frac{S_i(1)}{S_i(0)}$, $i = 1,2$ are the log returns of the portfolio and of stock i over time period $[0, 1]$.

Proof: Exercise.

Def 4.2

For an asset X, the expected return is defined by $\mu_X := \mathbb{E}[K_X]$.

E.g., $\mu_i := \mathbb{E}[K_i]$ for stock $i = 1, 2$, $\mu_V := \mathbb{E}[K_V]$ for the portfolio V.

The risk is defined as the standard deviation of the return:

$$\sigma_V := \sqrt{\text{Var}(K_V)} \quad \text{the risk of the portfolio;}$$
$$\sigma_i := \sqrt{\text{Var}(K_i)} \quad \text{the risk of stock } i, \quad i = 1, 2.$$

For two assets X, Y, we define their correlation coefficient, denoted by ρ_{XY}, as the correlation between K_X and K_Y.

Rmk 4.2

Recall

$$\rho_{XY} := \frac{\text{Cov}(K_X, K_Y)}{\sigma_X \sigma_Y} = \frac{\mathbb{E}[(K_X - \mathbb{E}[K_X])(K_Y - \mathbb{E}[K_Y])]}{\sigma_X \sigma_Y},$$

when $\sigma_X > 0$, $\sigma_Y > 0$. Moreover, if $|\rho_{XY}| = 1$, then the returns K_X, K_Y are affinely related, *i.e.*, there exist α, β s.t.

$$K_X = \alpha K_Y + \beta, \quad \text{sign}(\alpha) = \text{sign}(\rho_{XY}).$$

Prop 4.3

For a portfolio consisting of two securities,

$$\begin{aligned}
\mu_V &= w_1 \mu_1 + w_2 \mu_2, \\
\sigma_V^2 &= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \text{Cov}(K_1, K_2) \\
&= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \rho_{12} \sigma_1 \sigma_2.
\end{aligned}$$

Proof: Exercise

Eg 4.2

Use the following data to find the risk of the portfolio if $\mu_V = 5.96\%$.

Scenario	Probability	K_1	K_2
ω_1	0.3	-8%	15%
ω_2	0.7	10%	5%

We need to find the weights w_1, w_2. For that, first compute the expected returns:

$\mu_1 = 0.3 \cdot (-0.08) + 0.7 \cdot 0.10 = 0.046$ and $\mu_2 = 0.3 \cdot 0.15 + 0.7 \cdot 0.05 = 0.08$.

Then, in order to determine the respective weights, we solve the system

$$\begin{cases} w_1 + w_2 = 1 \\ w_1\mu_1 + w_2\mu_2 = \mu_V \end{cases} \Leftrightarrow \begin{cases} w_1 = 0.6 \\ w_2 = 0.4 \end{cases}.$$

We can compute the rest of the data: $\sigma_1^2 = 0.006804$, $\sigma_2^2 = 0.0021$, Cov$(K_1, K_2) = -0.00378$. Finally, we use the formula in Prop 4.3 to get

$$\sigma_V = \sqrt{w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\text{Cov}(K_1, K_2)} = \sqrt{0.00097104} = 0.03116.$$

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Prop 4.4

If short selling are not allowed, then $\sigma_V \leq \max\{\sigma_1, \sigma_2\}$.

Proof: ☼ No short selling means $x_1, x_2 \geq 0$, namely $w_1, w_2 \geq 0$. Assume, WLOG, that $\sigma_1 \geq \sigma_2$ or equivalently $\sigma_1 = \max\{\sigma_1, \sigma_2\}$. Then,

$$\begin{aligned}
\sigma_V^2 &= w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\sigma_1\sigma_2\rho_{12} \\
&\leq w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\sigma_1\sigma_2 \\
&= (w_1\sigma_1 + w_2\sigma_2)^2 \\
&\leq (w_1\sigma_1 + w_2\sigma_1)^2 \\
&= \sigma_1^2 = \max\{\sigma_1^2, \sigma_2^2\}.
\end{aligned}$$

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Risk Reduction (minimize σ_V) - The extreme cases

Note that $\rho_{12} \in [-1, 1]$. First consider the risk reduction in two extreme cases.

Prop 4.5

(a) If $\rho_{12} = 1$, then $\sigma_V = |w_1\sigma_1 + w_2\sigma_2|$. In particular, when $\sigma_1 \neq \sigma_2$,

$$\sigma_V = 0 \quad \text{iff} \quad w_1 = \frac{-\sigma_2}{\sigma_1 - \sigma_2} \text{ and } w_2 = \frac{\sigma_1}{\sigma_1 - \sigma_2}$$

Here either w_1 or w_2 is negative $\Rightarrow$ short sell must happen.

(b) If $\rho_{12} = -1$, then $\sigma_V = |w_1\sigma_1 - w_2\sigma_2|$. In particular,

$$\sigma_V = 0 \quad \text{iff} \quad w_1 = \frac{\sigma_2}{\sigma_1 + \sigma_2} \text{ and } w_2 = \frac{\sigma_1}{\sigma_1 + \sigma_2}.$$

Note that both weights are positive $\Rightarrow$ no short sell are necessary.

Proof:

(a) Since $\rho_{12} = 1$, we get

$$\sigma_V^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\sigma_1\sigma_2 = (w_1\sigma_1 + w_2\sigma_2)^2.$$

So $\sigma_V = |w_1\sigma_1 + w_2\sigma_2|$.

Risk Reduction (minimize σ_V) - The extreme cases

Proof continued:

(a) Since $\sigma_1 \neq \sigma_2$, we get $\sigma_V = 0$ if and only if

$$\begin{cases} w_1 + w_2 = 1 \\ w_1\sigma_1 + w_2\sigma_2 = 0 \end{cases} \Leftrightarrow \begin{cases} w_1 = -\frac{\sigma_2}{\sigma_1 - \sigma_2} \\ w_2 = \frac{\sigma_1}{\sigma_1 - \sigma_2} \end{cases}$$

(b) Analogous computation leads to the result; left as exercise.

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Def 4.3

Let A, B be two points on the plane.

(a) Every point on the plane that can be written as

$$xA + (1-x)B \quad \text{for some } x \in \mathbb{R},$$

is called affine combination of A and B.

(b) Every point on the plane that can be written as

$$xA + (1-x)B \quad \text{for some } x \in [0,1],$$

is called convex combination of A and B.

Risk Reduction (minimize σ_V) - The extreme cases

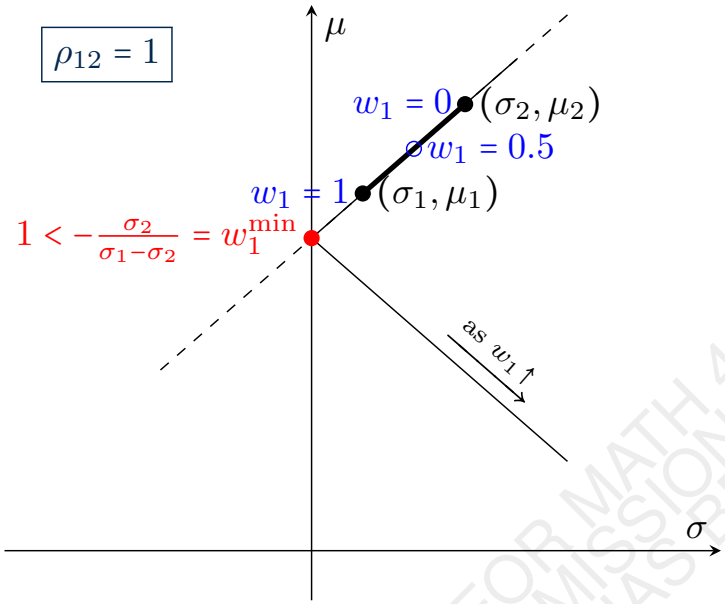

Figure: Positive correlation

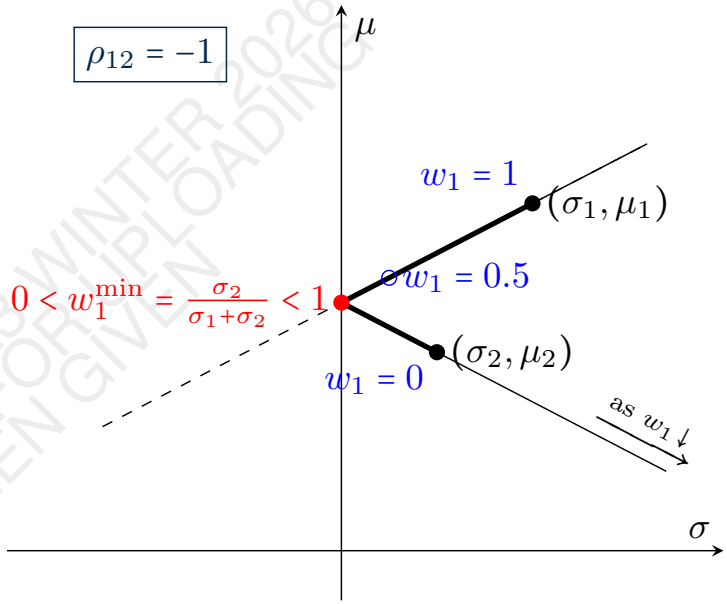

Figure: Negative correlation

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Risk Reduction (minimize σ_V) - The case $-1 < \rho_{12} < 1$

Prop 4.6

For $-1 < \rho_{12} < 1$, the minimal risk is attained when

$$w_1 = w_1^{\min} := \frac{\sigma_2^2 - \rho_{12}\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12}\sigma_1\sigma_2},$$

$$w_2 = w_2^{\min} := 1 - w_1^{\min} = \frac{\sigma_1^2 - \rho_{12}\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12}\sigma_1\sigma_2}.$$

If short selling are not allowed, the smallest risk (may not be 0) is attained at

$$(w_1, w_2) = \begin{cases} (0,1) & \text{if } w_1^{\min} < 0 \\ (w_1^{\min}, w_2^{\min}) & \text{if } 0 \le w_1^{\min} \le 1 \\ (1,0) & \text{if } 1 < w_1^{\min} \end{cases}$$

Proof: View σ_V^2 as a function of w_1:

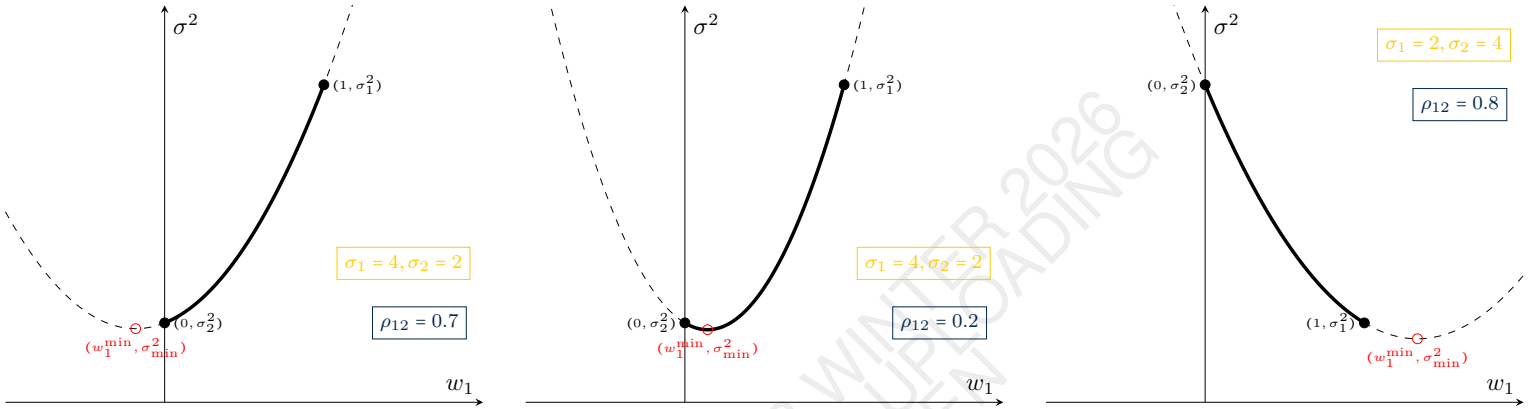
$$\sigma_V^2 = w_1^2\sigma_1^2 + (1-w_1)^2\sigma_2^2 + 2w_1(1-w_1)\rho_{12}\sigma_1\sigma_2 =: f(w_1).$$

This is a quadratic function. We can verify that $f'(w_1^{\min}) = 0$, $f''(w_1^{\min}) > 0$. So the minimal risk is attained at $(w_1^{\min}, w_2^{\min})$.

The cases where no short selling is allowed are immediate.

Risk Reduction (minimize σ_V)

Proof continued:



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Corollary (Cor) 4.1

Suppose $\sigma_1 \leq \sigma_2$, then we summarize the different cases as follows:

Cases	Is there a portfolio with $\sigma_V < \sigma_1 (\leq \sigma_2)$?	short selling needed for such a portfolio?
1) $-1 \leq \rho_{12} < \frac{\sigma_1}{\sigma_2}$	yes	no
2) $\rho_{12} = \frac{\sigma_1}{\sigma_2}$	no	not applicable
3) $\frac{\sigma_1}{\sigma_2} < \rho_{12} \leq 1$	yes	yes

Proof: See Capinski Cor 3.20

Risk Reduction (minimize σ_V)

Exercise

Determine the risk of a portfolio consisting of a risky asset with data (σ, μ) and a risk-free asset with return r.

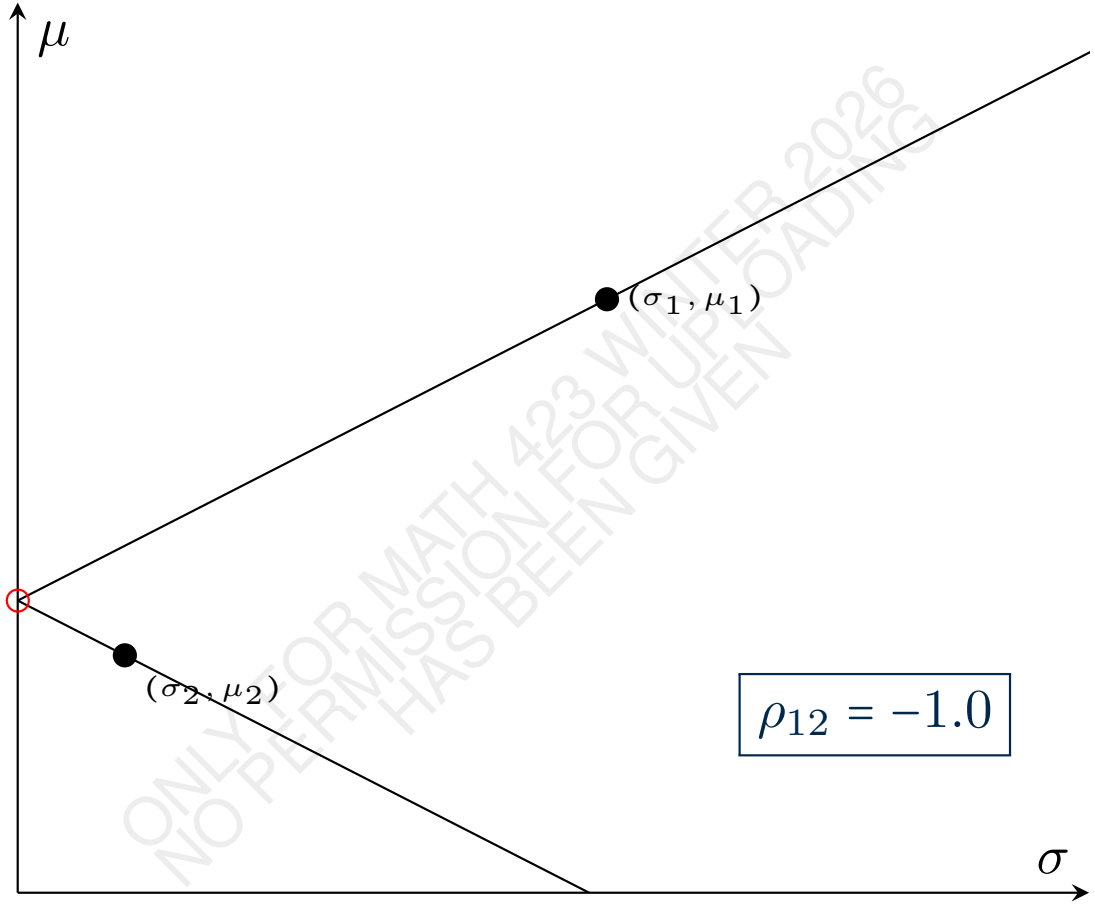
Def 4.4

The collection of all portfolios that can be constructed by investing in given assets is called the feasible (or attainable) set.

Rmk 4.3

When a portfolio consists of two assets, each point on the feasible set can be uniquely determined by the pair (w_1, w_2).

Risk Reduction (minimize σ_V) - The feasible set when ρ_{12} varies



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